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Revised Diagnostic Criteria for Vascular Cognitive Impairment and Dementia—The VasCog-2-WSO Criteria

The VasCog-2-WSO Criteria Consortium

IMPORTANCE Several sets of diagnostic criteria have been proposed for vascular cognitive impairment and dementia (VCID). The International Society for Vascular Behavioural and Cognitive Disorders (VasCog) working group published comprehensive operationalized criteria in 2014. Considering subsequent advances in the field, a revision was needed.

OBJECTIVE To update the VasCog criteria to achieve consensus on diagnosis of VCID.

DESIGN, SETTING, AND PARTICIPANTS VasCog criteria and other published diagnostic guidelines, aided by literature review of recent developments in VCID, were used as reference points for an online Delphi survey (minimum 3 rounds, ≥75% threshold for agreement), including operationalization of criteria and guidance on potential biomarkers. Seventy international experts from diverse international regions were invited to participate in 2023.

RESULTS Three survey rounds included 49 to 54 participants that agreed on VasCog-2 diagnostic criteria for preclinical, mild, and major dementia levels of vascular cognitive impairment (under the overarching term *VCID*). Research guidelines, including the use of novel neuroimaging and fluid biomarkers, were also agreed on. The World Stroke Organization (WSO) endorsed the criteria, hence named *VasCog-2-WSO*.

CONCLUSIONS AND RELEVANCE The VasCog-2-WSO criteria update the VasCog criteria for the diagnosis of VCID, providing operationalization and additional guidance on potential neuroimaging and fluid biomarkers. VasCog-2-WSO should provide an international standard for VCID diagnosis, facilitating diagnostic consistency among clinicians and researchers.

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Supplemental content

Group Information: The authors of the VasCog-2-WSO Criteria Consortium appear at the end of the article.

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Vascular dementia (VaD) is the second most common form of dementia, accounting for 15% to 20% of cases[1-3] and cerebrovascular disease (CeVD) may contribute to cognitive and functional decline in up to 75% of all persons with dementia.[4] Despite its importance and the promise of prevention, the advancement of research and clinical practice in this field has been limited. One major obstacle is the absence of universally recognized and clearly operationalizable diagnostic criteria, reducing the validity and interpretability of clinical outcomes and research findings.

The introduction of the National Institute of Neurological Disorders and Stroke-Association Internationale pour la Recherche et l'Enseignement en Neurosciences vascular dementia criteria[5] in 1993 was an important step toward standardization. This allowed major investigations on the natural history and therapeutics to be undertaken. Thereafter, multiple attempts have been made to enhance these criteria. For instance, the expert consensus criteria developed by the American Heart Association and American Stroke Association (AHA-ASA) in 2011,[1] the International Society of Vascular Behavioural and Cognitive Disorders (VasCog) criteria[6] (hereafter, *VasCog-1* criteria) in 2014 that coevolved with the *Diagnostic and Statistical Manual of Mental Disorders* (*Fifth Edition*; *DSM-5*) criteria released in 2013,[7,8] the Vascular Impairment of Cognition Classification Consensus Study (VICCCS-2) criteria of 2018, derived via a Delphi consensus process,[9,10] and the *International Classification of Diseases, 11th Revision* (*ICD-11*) criteria proposed in 2019.[11]

A study in 2019[12] showed that the VasCog-1 criteria had greater sensitivity and better predictive validity than the older criteria for vascular cognitive impairment and dementia (VCID) but were comparable with the *DSM-5* and VICCCS criteria. VasCog-1 has advantages over other criteria, such as more detailed operationalization (eg, suggested thresholds for cognitive impairment, severity of neuroimaging abnormalities, etc). Despite the subsequent release of VICCCS-2 and *ICD-11* criteria, there remains an ongoing lack of consensus on the overarching terminology of the cognitive decline, differing guidelines for clinical and cognitive assessments, and lack of accounting for major advances in neuroimaging and biomarkers. First, concerning terminology, the construct of VaD was broadened to the spectrum of VCI[13] that included individuals at increased risk of dementia as well as those with mild cognitive impairment (MCI) or dementia. Other terms proposed were vascular cognitive disorder,[6,14] VCID,[1] vascular neurocognitive disorder (*DSM-5*),[7] and dementia due to CeVD.[11] This

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lack of consensus on terminology has prevented consistency in definitions and diagnostic criteria.

Second, guidelines for assessment vary across the different criteria. The VICCCS-2 and AHA-ASA criteria do not require concern of a person, informant, or clinician regarding decline of cognitive functioning for diagnosis, whereas VasCog-1 and *DSM-5* criteria require this in alignment with the criteria for Alzheimer disease (AD).[15] Most criteria specify executive function, attention, memory, language, and visuospatial function as the cognitive domains of interest and most likely to be affected by cerebrovascular lesions.[16] The VICCCS-2 criteria, on the other hand, indicate processing speed, learning, and neuropsychiatric symptoms[17] as optional domains. Similarly, specificity of symptoms listed as supportive evidence in VasCog-1 criteria, including early urinary frequency, gait disturbance, and personality and mood changes, may be limited.

Third, there have been major advances in neuroimaging and the development of other biomarkers for some types of dementia, and this needs to be revisited in the context of VCID. Consensus neuroimaging standards for small vessel disease have been published,[16,17] and their incorporation in any criteria set is arguably important. There have also been advances in the genetics of VCID that may help guide a phenotype of relatively pure VCID.[18]

Fourth, the status of VCID subtypes, with terms such as poststroke dementia, subcortical ischemic vascular dementia, and multi-infarct dementia still being commonly used by clinicians, requires clarification within the nosology of VCID.

This review, therefore, aimed to reach international consensus on the diagnosis of VCID, via a revision of the VasCog-1 criteria, in an iterative survey using a Delphi approach.

Key Points

Question Can consensus be achieved for updated diagnostic criteria for vascular cognitive impairment and dementia (VCID) accounting for recent research developments?

Findings In this Delphi study including 70 invited participants, 49 to 54 international experts agreed on diagnostic criteria for preclinical, mild, and major dementia levels of vascular cognitive impairment (under the overarching term *VCID*). Biomarkers were incorporated into criteria primarily as features that may suggest an alternative or concomitant etiology to vascular disease.

Meaning The VasCog-2-WSO criteria provide operationalized guidance that accounts for recent research advancements and should provide an international standard for VCID diagnosis.

Methods

Participants

This review was approved by the University of New South Wales Human Research and Ethics Committee. Participants gave written informed consent. Survey participants were invited if they were internationally recognized clinicians and/or researchers in the field of cognitive decline and dementia with a focus on VCID. These could be medical specialists (including geriatricians, neurologists, psychiatrists, and rehabilitation physicians) or nonmedical clinician experts (eg, neuropsychologist, nurse, social worker, occupational therapist). Experts who had previously participated in the VasCog-1, VICCCS-2, and AHA-ASA criteria were initially contacted, and snowball sampling was used to expand this group. The aim was to have broad representation from all geographical regions including South and East Asia, North and South America, Africa, Europe, and Oceania.

A total of 310 individuals were initially screened for invitation, of whom 143 were short-listed. Of these, 70 responded indicating their interest in participation (9 declined, the remainder did not respond). The World Stroke Organization (WSO) was approached to provide input into the process, and 2 members of WSO with expertise in vascular dementia participated.

Data Collection

A modified Delphi method[19] was used, involving an iterative, multistaged series of structured online questionnaires with anonymized feedback from participants, and progressive refinement of questions across stages to reach consensus for the updated diagnostic criteria. Only the independent coordinator (A.C.B., who did not participate in the survey) had access to identification details of the participants. The Guidance on Conducting and Reporting of Delphi Studies (CREDES) recommendations were followed.[20] The **Figure** illustrates the procedure.

Development of the Survey

Literature reviews on topics related to the criteria were conducted by the core team (A.C.B., P.S.S., N.A.K., J.J., S.H., R.K., R.J.C., and D.S.), which informed the first structured questionnaire. In brief, this entailed consulting all previous diagnostic criteria for VCID, and major reviews on neuropsychological, neuroimaging, and biomarker developments in VCID since VasCog-1. Consensus was sought on the following:

- Each individual element of the existing VasCog criteria
- The standardization of definitions, terminology, and scope of the criteria
- Further operationalization of the criteria, and
- Potential additions to the criteria including novel neuroimaging, fluid, or retinal biomarkers

A full summary of the topics addressed in each round is provided (eTable 1 in the Supplement).

Survey and Consensus Procedure

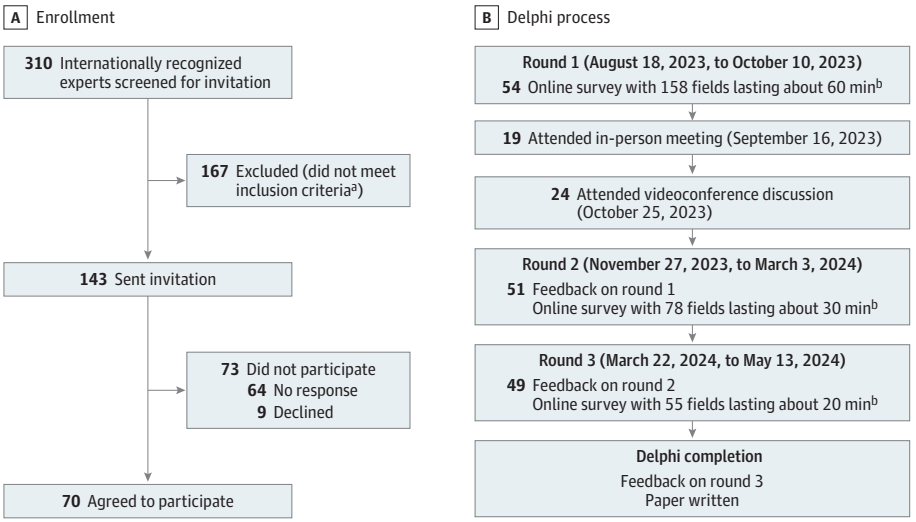
Consensus on any statement was defined as agreement or disagreement of greater than or equal to 75% and nonconsensus as less than 75% through multiple iterative rounds. Percentage agreement is the most common measure of agreement used in Delphi studies, and 75% is the median value used[21] and was considered high enough to ensure the quality and reliability of the consensus outcomes. Statements that reached consensus were removed from subsequent rounds, and nonconsensus statements were reworded incorporating participant feedback and reintroduced in subsequent rounds (Figure).

Overall, 3 rounds of online surveys were administered, approximately 1 every 2 months. In response to feedback, an in-person participants' meeting was held on September 16, 2023,

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Figure. Delphi Consensus Process Used for Defining the Diagnostic Criteria for Vascular Cognitive Impairment and Dementia (VCID)



[a]Selection criteria are as follows: (1) international experts in the neuropsychological assessment of VCID, recognized from the content of their international, peer-reviewed publications and/or referral by peers, (2) clinicians, researchers, or both with experience in the diagnosis of cognitive decline and dementia in older adults, and (3) medical specialists (including geriatricians, neurologists, psychiatrists, and rehabilitation physicians), but nonmedical clinician experts (eg, neuropsychologist, nurse, social worker, occupational therapist) were also included if they satisfied criteria (1) and (2).

[b]eTable 1 in the Supplement contains details of main topics per round.

at the VasCog 2023 conference in Gothenburg, Sweden, to discuss the round 1 results. VasCog encompasses clinician and researchers experts in cognitive disorders arising from cerebrovascular pathological changes. An additional meeting was held on October 25, 2023, by videoconference for those who could not attend VasCog 2023 to discuss contentious round 1 results and guide preparation of questions for round 2. No questionnaire data was collected during these meetings, preserving the anonymity of the process.

Observations

Primary Outcomes

Of the 70 participants invited, 54 (77%) contributed to round 1, 51 to round 2 (94% of round 1), and 49 to round 3 (97% of round 2). Characteristics of the Delphi participants from round 1 are shown in eTable 2 in the Supplement. In brief, the majority (89%) of participants were medically trained, and 6 (11%) were neuropsychologists; among the medically trained, most (77%) were neurologists, psychiatrists, geriatricians, or epidemiologists. Most reported their primary workplace setting as a university, university clinic, hospital, memory clinic, or stroke unit or some combination. Most reported having both clinical and research experience with VCID and having expertise demonstrated by relevant peer-reviewed research publications. The majority (78%) were from Europe, North America, or Australia with 5 (9%) from Asia, 2 (4%) from Brazil, and 1 (2%) from the Caribbean.

Definition of the Vascular Cognitive Construct

In round 1, the naming of the overarching vascular cognitive construct that includes all forms of cognitive deficits from mild to major dementia resulted in a split between the most popular options *vascular cognitive disorder* (30% agree) and *vascular cognitive impairment* (26%), with other options being *vascular neurocognitive disorder* (18%) and *vascular cognitive impairment and dementia* (16%). The question was reduced to 2 options in round 2, ie, *vascular cognitive impairment* (61%) and *vascular cognitive disorder* (39%), but subsequent discussion in a conference suggested that this was an unsatisfactory outcome, and the options were brought to round 3. In the round, the proposal to name the construct *VCID*, with the term *vascular MCI (vaMCI)* for a mild level of severity of VCID and the term *VaD* for a major or dementia-level of severity of VCID reached consensus (84%), and VCID, vaMCI, and VaD are therefore the default terms used hereafter.

Box 1 presents the diagnostic criteria for MCI and dementia reached by consensus. This is the first step in the diagnostic process, before establishing a vascular etiology. These criteria are little changed from VasCog-1 besides minor rewording.

The **Table** displays the consensus cognitive domains assessed in VCID, for which the *DSM-5* domains replace the VasCog-1 domains (80% in round 2) with minor changes, ie, changing the domain name *complex attention* to *attention and processing speed* and subsuming *praxis-gnosis-body schema* into *perceptual-motor function*.

The criteria for establishing vascular etiology for MCI or dementia are presented in **Box 2**.[18,22-29] For criterion A.1, in a separate question, consensus was not reached on the choice of a specific time frame (3 months, 6 months, other, or none) for the persistence of cognitive deficits after a stroke (3 months: 39% agreement; 6 months: 30%, round 2). Consequently, the beyond 3 months guidance was retained from VasCog-1 as the most supported option.

Criterion B.1 combines 2 criteria from VasCog-1, ie, former B.1 large vessel infarcts and B.2 extensive or strategically placed single infarct to simplify the criteria as former B.1 was not supported (56% disagreement). Otherwise, imaging criteria are marginally more lenient with respect to lacunes and white matter hyperintensities (Box 2).[18,22-29] For criterion A.2, only the supportive clinical features of personality and mood

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Box 1. Consensus Diagnostic Criteria for Mild Cognitive Impairment and Dementia

Mild Cognitive Impairment (Both A and B Are Necessary)

A. Acquired decline from a documented or inferred previous level of performance in one or more cognitive domains as evidenced by the following:
   1. Concerns of the person, knowledgeable informant or a clinician of mild levels of decline from a previous level of cognitive functioning.

   Guidance: Typically, the reports will involve greater difficulty in performing the tasks and greater effort, compensatory strategies, or accommodation may be required.

   1. Evidence of modest deficits on objective cognitive assessment based on a validated measure of neurocognitive function (either formal neuropsychological testing or an equivalent clinical evaluation[a]) in ≥1 cognitive domains listed in Box 2.

   Guidance: The test performance is typically in the range between 1 and 2 SDs below appropriate norms[b] (or between the 3rd and 16th percentiles) and/or below an estimate of the person's premorbid cognitive functioning when a formal neuropsychological assessment is available, or an equivalent level as judged by the clinician including from appropriate score cutoffs and/or decline measured from cognitive screening tests.

B. The cognitive deficits are not sufficient to interfere with independence (ie, instrumental activities of daily living are preserved), but greater effort, compensatory strategies, or accommodation may be required to maintain independence.

Dementia (Both A and B Are Necessary)

A. Evidence of substantial cognitive decline from a documented or inferred previous level of performance in one or more of the domains outlined above. Evidence for decline is based on (both A and B):
   1. Concerns of the person, a knowledgeable informant, or the clinician, of significant decline in specific abilities;
   2. Clear and significant deficits in objective assessment based on a validated objective measure of cognitive function (either formal neuropsychological testing or equivalent clinical evaluation[a]) in ≥1 cognitive domain.

   Guidance: Test performance typically falls ≥2 SDs below appropriate norms[b] (or below the third percentile) and/or below an estimate of the person's premorbid cognitive functioning when a formal neuropsychological assessment is available, or an equivalent level as judged by the clinician including from appropriate score cutoffs and/or decline measured from cognitive screening tests.

B. The cognitive deficits, in isolation from physical deficits, are sufficient to interfere with independence (eg, at a minimum requiring assistance with instrumental activities of daily living; ie, performing tasks such as managing finances or medications).

[a] The equivalent clinical evaluation should include objective and valid tests including cognitive screening tests but can replace a formal neuropsychological assessment if the latter is not available, except in research studies where a formal assessment is warranted.

[b] Appropriate norms should include, where possible, normative sample of similar age, sex, education, and sociocultural background.

changes reached consensus, unlike gait disturbance and urinary frequency, which were considered too nonspecific to be included within the diagnostic criteria.

Table. Consensus Cognitive Domains Assessed in Vascular Cognitive Impairment and Dementia

Cognitive domains	Subdomains/description
1. Attention and processing speed	Sustained attention, divided attention, selective attention, processing speed
2. Executive function	Planning, decision-making, working memory, responding to feedback, inhibition, flexibility
3. Learning and memory	Free recall, cued recall, recognition memory, semantic and autobiographical long-term memory, implicit learning
4. Language	Object naming, word finding, fluency, grammar and syntax, receptive language
5. Perceptual-motor function	Visual perception, visuoconstructional reasoning, perceptual-motor coordination
6. Social cognition	Recognition of emotions, theory of mind, insight

In the exclusion criteria, major depression was removed due to a lack of consensus on the original or reworded alternatives, and the agreement that there is considerable overlap between depression and VCID symptoms in later life. Mood disturbance, personality change, and apathy are included as supportive, but not essential, features for the diagnosis of VCID. The addition of biomarkers for AD and dementia with Lewy bodies (guidance) reached consensus but not for frontotemporal dementia primarily due to lack of validated biomarkers. There was consensus against (≥75% disagreement, round 1) the use of additional novel neuroimaging markers, plasma and serum, and retinal biomarkers in either the clinical or research setting to assist in VCID diagnosis (the eResults in the Supplement contains the list of markers). Consensus was reached for using neuroimaging markers for the staging of cerebrovascular disease in a research setting (81%, round 1). Details of this will be explored in a future research framework.

Box 3 presents miscellaneous criteria. For a diagnosis of probable VCID, a clinical diagnosis should be supported by neuroimaging or the presence of a genetic disorder such as cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy, the latter being a new inclusion. Another notable addition is the inclusion of the preclinical/at-risk VCID category, indicated by neuroimaging evidence of CeVD, with or without clinical history of cerebrovascular events and/or vascular risk factors. This recognizes the common occurrence of what has also been referred to by clinicians as covert CeVD.[30]

Multiple causation was simplified into vascular with neurodegenerative and nonneurodegenerative additional pathologies with the more clinically salient etiology to be stated. In VasCog-1, multiple causation was centered on AD. The subcortical/cortical-subcortical subtype from VasCog did not reach consensus, nor did a suggested poststroke dementia or poststroke VCID subtype, and the hemorrhagic/ischemic subtype was maintained with an added mixed hemorrhagic-ischemic category.

eTable 3 in the Supplement contains a summary of the VasCog-2-WSO diagnostic criteria for VCID, and eTable 4 in the Supplement displays comparison of VasCog-2-WSO criteria with VICCCS-2, VasCog-1, *DSM-5*, and AHA-ASA criteria.

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Box 2. Consensus Evidence for Predominantly Vascular Etiology of Cognitive Impairment

Both A and B are required. Features under (C) suggest multiple contributing etiologies (mixed or multifactorial cognitive impairment and dementia) or an alternative etiology, in which case the likely predominant etiology should be recognized.

A. One of the Following 2 Clinical Features (A.1. or A.2.)

1. The onset of the cognitive deficits is temporally related to ≥1 clinical strokes.
   Guidance: Onset is abrupt and cognitive deficits persist beyond 3 months after the stroke. Multiple strokes may be associated with a stepwise or fluctuating course.
   The evidence of stroke is 1 of the following:
   a. Documented history of a stroke, with cognitive decline temporally associated with the event.
   b. Physical signs consistent with stroke in the absence of a history of stroke or if the history of stroke is unknown, and the signs are not due to another cause including another neurodegenerative disorder.
   Guidance: Physical signs may include but are not limited to hemiparesis, lower facial weakness, Babinski sign, pronator drift, sensory deficit, visual field defect, pseudobulbar syndrome—supranuclear weakness of muscles of face, tongue and pharynx, spastic dysarthria, swallowing difficulties—and cerebellar signs eg, limb ataxia. Covert neuroimaging evidence (eg, brain infarct or hemorrhage) of cerebrovascular disease (see part B) may be present in the absence of clinical history of stroke and should be taken into consideration, as for physical signs of stroke.
2. In the absence of history of a stroke or transient ischemic attack, evidence for decline from subcortical ischemic pathology may be associated with a picture of gradual onset and slowly progressive course, typically predominant in some combination of attention and processing speed, and/or executive functioning, with additional domains potentially affected to a lesser extent.
   Guidance: Memory impairment, where present, typically involves inefficient encoding and/or retrieval (vs consolidation and storage in Alzheimer disease [AD]).
   Guidance (supportive features): Personality and mood changes, particularly apathy, depressive symptoms and emotional lability, and gait-balance disorders are common in vascular cognitive impairment and dementia.

B. Presence of Significant Neuroimaging Magnetic Resonance Imaging (MRI) (Preferable) or Computed Tomography (CT) Evidence of Cerebrovascular Disease (at Least 1 of the Following)

1. Multiple infarcts or a single extensive or strategically placed infarct, for example in the thalamus, may be sufficient for vascular dementia.
2. Multiple lacunes (≥2) outside the brainstem; 1 lacune may be sufficient if strategically placed or in combination with extensive white matter hyperintensities.
3. White matter hyperintensities, particularly if they are extensive and confluent.
   Guidance: Extensive and confluent may correspond to a Fazekas score[22] of 2 or 3; or by region: periventricular Fazekas = 3 and/or deep Fazekas = 2 or 3.
4. Intracerebral hemorrhage; one may be sufficient if large and/or in a lobar location or otherwise strategically placed, or two or more intracerebral hemorrhages.

(continued)

Box 2. (continued)

Guidance (supportive features for cerebrovascular burden, not sufficient on their own without meeting neuroimaging criteria above):
   a. Cerebral microbleeds and cortical superficial siderosis
   b. Abnormalities that meet the criteria of cerebral amyloid angiopathy (eg, Boston criteria version 2.0[23])
Guidance (all imaging evidence): The Standards for Reporting Vascular Changes on Neuroimaging 2 (STRIVE-2)[18] definitions and guidelines of measurement of cerebrovascular lesions and changes should be adopted.

C. Features That May Suggest an Alternative Predominant Etiology or That Vascular Disease Is Not the Sole Etiology

1. Clinical features
   a. Insidious early onset of cognitive, perceptual and motor symptoms suggestive of AD in the absence of corresponding focal vascular lesions (infarct or small vessel disease) on brain imaging or history of vascular events
   Guidance: Where available, supported by indicative biomarkers as outlined in 3a below.
   a. Early and prominent movement disorder suggestive of dementia with Lewy bodies or other α-synucleinopathy or other nonvascular movement disorder
   Guidance: Where available, supported by indicative biomarkers (as per[24] for dementia and[25] for mild cognitive impairment) and outlined in 3b below.
   a. Features strongly suggestive of another primary neurological disorder such as multiple sclerosis, encephalitis, toxic or metabolic disorder, etc. sufficient to explain the cognitive impairment
2. Neuroimaging
   a. Absent or minimal cerebrovascular lesions on CT or MRI
3. Biomarkers
   a. The presence of any of the following biomarkers for AD may provide supportive evidence for AD as an alternative cause of the cognitive syndrome, or that AD is present in addition to vascular disease.
      i. Cerebrospinal or plasma amyloid β (Aβ) and phosphorylated tau (p-tau)–derived species such as Aβ42:40 ratio, p-tau181, p-tau 217 or other established biomarker[26,27] at levels consistent with brain amyloid pathology
      ii. Imaging brain amyloid levels (ie, using positron emission tomography) at accepted Centiloid/standardized uptake value ratio thresholds
      iii. The presence of an autosomal mutation associated with AD, or being homozygous for apolipoprotein E ε4.[a]
   b. The presence of biomarkers of dementia with Lewy bodies,[25] including dopamine transporter imaging and α-synuclein seed amplification assay in cerebrospinal fluid.[28a]
   c. The presence of biomarkers for any other disease process known to be related to cognitive decline and judged to be implicated (eg, pathogenetic variant of frontotemporal dementia,[29] neuroimaging evidence of brain tumor or traumatic brain injury, etc).[a]

[a] Criteria 3aiii, 3b, and 3c were not part of the consensus Delphi process, and were added during preparation of this paper as considered by the group to add important details concerning biomarkers for potential differential diagnoses.

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Box 3. Consensus Criteria for Vascular Cognitive Impairment and Dementia (VCID; Vascular Mild Cognitive Impairment or Vascular Dementia): Miscellaneous Aspects

Level of Certainty

1. Probable (a or b are required)
   a. Clinical criteria for VCID are supported by neuroimaging
   b. Both clinical and genetic evidence of cerebrovascular diseases.
   Guidance: Genetic disorders include but are not limited to cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL); cerebral autosomal recessive arteriopathy with subcortical autosomal recessive leukoencephalopathy (CARASIL); hereditary endotheliopathy retinopathy nephropathy and stroke (HERNS); pontine autosomal dominant microangiopathy and leukoencephalopathy (PADMAL); retinal vasculopathy with cerebral leukodystrophy (RVCL); and collagen type IV, α-1 (COL4A1)–related disorders.
2. Possible. Clinical criteria for VCID are met, but neuroimaging and/or genetic evidence is not available

Subtypes of VCID

Hemorrhagic, ischemic, or mixed hemorrhagic-ischemic.

Multiple or Mixed Causation (Multifactorial or Mixed Cognitive Impairment or Dementia)

State which etiology is clinically more salient: vascular or other.

1. VCID with 1 or more concomitant neurodegenerative disease(s) (eg, Alzheimer disease [AD], dementia with Lewy bodies) (a and b are required)
   a. Meets criteria for VCID (except for features that suggest an alternative etiology/etiologies)
   b. Meets criteria for other neurodegenerative disease(s)
   c. (eg, VCID [primary] with contribution from AD; or AD [primary] with contribution from VCID)
2. VCID with additional nonneurodegenerative pathology (eg, VCID [primary] with contribution from traumatic brain injury, brain tumor, etc)
3. VCID with contribution from depression
   Associated behavioral or psychiatric symptoms: with apathy, depressive symptoms, emotional lability, psychotic symptoms, agitation, etc.

Preclinical/At-Risk VCID

The purpose of this group would be to signal the importance of further study, primarily in a research setting, and for risk reduction and preventive medicine. The category would require the person to present with (a and b):

a. Significant incidental neuroimaging evidence (magnetic resonance imaging [preferred] or computed tomography) of cerebrovascular disease that satisfies the VCID criteria for vascular etiology criterion B; and
b. Does not satisfy criteria for mild or major cognitive impairment/dementia and the person, their informant or relevant clinician do not have concerns of cognitive change
   Guidance: this diagnosis is supported by the presence of substantial vascular risk factors for VCID.

Discussion

This review provides consensus VasCog-2-WSO diagnostic criteria for VCID. These criteria aim to update the VasCog-1 criteria to address inconsistencies between extant criteria in terminology and assessment guidelines and align with advances in neuroimaging and biomarkers.[6] The overarching vascular concept was named VCID, maintaining its subdivision into vaMCI and VaD. The primary updates include refinement of the VasCog neuroimaging and clinical criteria and subtyping, the addition of AD and other neurodegenerative disease biomarkers in criteria for alternative or additional etiologies, and the addition of a preclinical/at-risk category for risk reduction and preventive purposes. These updates will be of great value in reflecting the substantial advancement of knowledge since VasCog-1, addressing limitations of the previous criteria, and thereby facilitate ongoing clinical and research developments.

The VasCog-2-WSO are highly operationalized, with needed, practical guidance on thresholds for cognitive and neuroimaging criteria. This is designed to assist with decision-making in both clinical and research settings and remain sufficiently broad and flexible to facilitate clinical judgment. For example, criteria for white matter hyperintensities (WMHs) include guidance on Fazekas score thresholds[31] in recognition that WMHs are common and increase with age[32] and that visual scoring methods remain in use in the clinic. For other neuroimaging markers (ie, infarcts, lacunes, and intracerebral hemorrhage) there is reference to strategic placement in recognition of the long-understood associations between lesion location and the risk of cognitive impairment (eg, thalamus and basal ganglia) with recent lesion-mapping studies enhancing our understanding[22,33,34].

VasCog-2-WSO, like the VasCog-1 criteria, recognize that the neuropsychological domains of attention and processing speed (within complex attention in VasCog) and executive function are affected early in the disease process and are helpful in identifying small vessel disease–related impairment in the early stages.[35] The heterogeneity of the cognitive profile of VCID is also acknowledged, as multiple domains are commonly affected owing to the widespread nature of the cerebral pathology. This includes learning and memory, a feature commonly associated with AD. Memory impairment in early VCID, however, may present differently from early AD, with reduced efficiency and organization of learning and retrieval in the former, and impaired consolidation and rapid forgetting in the latter.[36] Tests such as a repeated supra-span word list, with measures of learning over trials, free recall and recognition can help distinguish the nature of the learning and memory deficits in the 2 disorders. The VasCog-2-WSO consensus cognitive domains are harmonized with the domains in the consensus VasCog-2–neuropsychological battery and standards.[37] Other supportive aspects of the clinical criteria considered insufficiently specific (eg, gait and/or urinary disturbance as supportive and major depression as exclusion criteria) were removed, which may improve diagnostic accuracy.

An important aspect of any dementia diagnosis is its level of certainty. The approach in the previous criteria sets for both VCID and AD has been to have 2 levels—probable and possible. In AD, the probable level is a clinical construct supported by molecular biomarkers and genetics.[38] Biomarkers are similarly important for probable VCID, but molecular biomarkers for VCID are currently lacking,[39] and neuroimaging, in particular MRI, remains the mainstay. There are several neuroimaging biomark-

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ers—large and small infarcts, WMHs, lacunes, dilated perivascular spaces, macrobleeds and microbleeds, cerebrovascular reactivity, diffusivity measures, and blood-brain barrier integrity measures. The definitions of the small vessel disease-related measures have been standardized in the Standards for Reporting Vascular Changes on Neuroimaging (STRIVE) criteria.[16,17] Several of the following challenges remain: not all of these markers (eg, some WMHs) are necessarily vascular in origin,[40] with the description of presumed vascular origin being used[17]; these lesions are commonly seen as incidental findings in cognitively healthy older people[41] and in neurodegenerative diseases including AD,[42] and the association between the severity of these markers and cognitive impairment is generally modest.[43] Further work in developing a biomarker profile for VCID is needed to place the diagnosis of probable VCID on a more secure footing. Although there is accumulating evidence of monogenic vascular disorders that lead to VCID, these are very uncommon, with most cases being sporadic, albeit with genetic risk factors.[44] The monogenic disorders can, however, serve as models of a pure form of VCID.[18]

The Delphi process of VasCog-2-WSO attempted the subtyping of VCID. Previous subtyping of VCID, such as into post-stroke dementia, subcortical ischemic vascular dementia, and multi-infarct dementia, were not supported by consensus, and the ischemic/hemorrhagic subtype was maintained with an added mixed category. Instead of intensive subtyping, the focus of VasCog-2-WSO criteria is on the evidence for vascular etiology per se, with evidence for stroke and nonstroke etiology included as separate criteria. Although a lack of consensus on these specific terms likely reflects their current inadequacy, future attempts at subtyping should not be ruled out.

The VasCog-2-WSO preclinical/at-risk VCID category is aimed at identifying nonsymptomatic persons with high levels of CeVD, discovered incidentally via neuroimaging, and hence at risk of cognitive decline and stroke that could be targeted for monitoring, risk reduction, and preventative interventions particularly in a research setting. Although this is similar to the brain-at-risk concept proposed several years ago,[39] the recent interest in covert CeVD and the possibility of interventions to slow its progression is noteworthy.[30]

VasCog-2-WSO present novel features that may suggest VCID may not be the primary or sole pathology, using AD biomarkers such as cerebrospinal fluid Aβ and p-tau levels and amyloid imaging using positron emission tomography, as well as biomarkers for other pathologies such as dementia with Lewy bodies (Box 3). However, these novel features should be used cautiously as the presence of these features or biomarkers does not exclude the possibility of multiple brain pathologies being present, as is common in older people.[4] The criteria therefore encourage the identification of multiple pathologies and support an attempt to determine the most salient pathologies that possibly underlie cognitive impairment. This is important for identifying vascular pathologies among multiple pathologies in relation to the selection for current and future amyloid and other neurodegeneration focused therapies.[45]

Strengths of these criteria include their determination by Delphi consensus process that involved a broad international representation of experts with excellent engagement, with the process including 1 in-person meeting and a videoconference, and adherence to the CREDES recommendations.[20] The benchmark for all criteria, consensus at 75%, was high. The criteria were operationalized for use by both clinicians and researchers, including the determination of vascular etiology. This contrasts with most previous criteria that adopt more general guidelines. Hence, the VasCog-2-WSO criteria are arguably more useful for decision-making, more amenable to validation against biological criteria for CVD such as neuropathology, and provide a clearer framework for ascertaining prognosis and guiding treatment.[1,46,47] They are consistent with *DSM-5* criteria as well as the latest criteria for AD and dementia with Lewy bodies and their biomarkers, adopt the Standards for Reporting Vascular Changes on Neuroimaging 2 (STRIVE-2) standards, and align with the harmonized VasCog-2–neuropsychological assessment.[37] The combination of these features positions the VasCog-2-WSO criteria well for future research and development.

Like all exercises aiming to reach expert consensus, the criteria run the risk of prevalent biases that are only correctable with novel data. The criteria remain to be tested in the field, and their validation is an important future goal to build on previous efforts involving VasCog-1 and other criteria.[12] Despite its adoption here, it is recognized that terms such as *dementia* are potentially limited and not always preferred, and it should be interchangeable with alternative names such as *major cognitive impairment* or *major cognitive disorder*.

Future Directions

The VasCog-2-WSO should be validated against other extant criteria and in longitudinal studies for which neuropathological data are available. Their utility for participant selection in clinical trials needs to be determined. Their ease of application in the clinic and population level should be examined. The criteria may require some refinement for exclusive research use. However, real advances in diagnostic criteria will only come with better biomarkers, and this is an area of active research internationally, with novel neuroimaging, retinal and fluid biomarkers under investigation.

Conclusions

The VasCog-2-WSO criteria update the VasCog criteria for the diagnosis of VCID, providing operationalization and additional guidance on potential neuroimaging and fluid biomarkers. VasCog-2-WSO should provide an international standard for VCID diagnosis, facilitating diagnostic consistency among clinicians and researchers.

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REFERENCES

1. Gorelick PB, Scuteri A, Black SE, et al; American Heart Association Stroke Council, Council on Epidemiology and Prevention, Council on Cardiovascular Nursing, Council on Cardiovascular Radiology and Intervention, and Council on Cardiovascular Surgery and Anesthesia. Vascular contributions to cognitive impairment and dementia: a statement for health care professionals from the American Heart Association/American Stroke Association. *Stroke*. 2011;42(9):2672-2713. doi:10.1161/STR.0b013e3182299496

2. Lobo A, Launer LJ, Fratiglioni L, et al; Neurologic Diseases in the Elderly Research Group. Prevalence of dementia and major subtypes in Europe: a collaborative study of population-based cohorts. *Neurology*. 2000;54(11)(suppl 5):S4-S9.

3. Rockwood K, Wentzel C, Hachinski V, Hogan DB, MacKnight C, McDowell I; Vascular Cognitive Impairment Investigators of the Canadian Study of Health and Aging. Prevalence and outcomes of vascular cognitive impairment. *Neurology*. 2000; 54(2):447-451. doi:10.1212/WNL.54.2.447

4. Boyle PA, Wang T, Yu L, et al. To what degree is late life cognitive decline driven by age-related neuropathologies? *Brain*. 2021;144(7):2166-2175. doi:10.1093/brain/awab092

5. Román GC, Tatemichi TK, Erkinjuntti T, et al. Vascular dementia: diagnostic criteria for research studies. Report of the NINDS-AIREN International Workshop. *Neurology*. 1993;43(2):250-260. doi:10.1212/WNL.43.2.250

6. Sachdev P, Kalaria R, O'Brien J, et al; International Society for Vascular Behavioural and Cognitive Disorders. Diagnostic criteria for vascular cognitive disorders: a VASCOG statement. *Alzheimer Dis Assoc Disord*. 2014;28(3):206-218. doi:10.1097/WAD.0000000000000034

7. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders: DSM-5*. Vol 5. American Psychiatric Association; 2013.

8. Sachdev PS, Blacker D, Blazer DG, et al. Classifying neurocognitive disorders: the DSM-5 approach. *Nat Rev Neurol*. 2014;10(11):634-642. doi:10.1038/nrneurol.2014.181

9. Skrobot OA, Black SE, Chen C, et al; VICCCS group. Progress toward standardized diagnosis of vascular cognitive impairment: guidelines from the Vascular Impairment of Cognition Classification Consensus Study. *Alzheimers Dement*. 2018;14(3): 280-292. doi:10.1016/j.jalz.2017.09.007

10. Skrobot OA, O'Brien J, Black S, et al; VICCCS group. The Vascular Impairment of Cognition Classification Consensus Study. *Alzheimers Dement*. 2017;13(6):624-633. doi:10.1016/j.jalz.2016.10.007

11. World Health Organization. International Classification of Diseases, 11th revision. Accessed June 19, 2025. https://icd.who.int/en/

12. Sachdev PS, Lipnicki DM, Crawford JD, Brodaty H. The vascular behavioral and cognitive disorders criteria for vascular cognitive disorders: a validation

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study. *Eur J Neurol*. 2019;26(9):1161-1167. doi:10.1111/ene.13960

13. Hachinski V, Iadecola C, Petersen RC, et al. National Institute of Neurological Disorders and Stroke–Canadian Stroke Network vascular cognitive impairment harmonization standards. *Stroke*. 2006;37(9):2220-2241. doi:10.1161/01.STR.0000237236.88823.47

14. Román GC, Sachdev P, Royall DR, et al. Vascular cognitive disorder: a new diagnostic category updating vascular cognitive impairment and vascular dementia. *J Neurol Sci*. 2004;226(1-2):81-87. doi:10.1016/j.jns.2004.09.016

15. Albert MS, DeKosky ST, Dickson D, et al. The diagnosis of mild cognitive impairment due to Alzheimer disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer disease. *Alzheimers Dement*. 2011;7(3):270-279. doi:10.1016/j.jalz.2011.03.008

16. Duering M, Biessels GJ, Brodtmann A, et al. Neuroimaging standards for research into small vessel disease-advances since 2013. *Lancet Neurol*. 2023;22(7):602-618. doi:10.1016/S1474-4422(23)00131-X

17. Wardlaw JM, Smith EE, Biessels GJ, et al; Standards for Reporting Vascular Changes on Neuroimaging (STRIVE v1). Neuroimaging standards for research into small vessel disease and its contribution to ageing and neurodegeneration. *Lancet Neurol*. 2013;12(8):822-838. doi:10.1016/S1474-4422(13)70124-8

18. Saks DG, Sachdev PS. Monogenic causes of cerebral small vessel disease—models for vascular cognitive impairment and dementia? *Curr Opin Psychiatry*. 2025;38(2):112-118. doi:10.1097/YCO.0000000000000978

19. Graham B, Regehr G, Wright JG. Delphi as a method to establish consensus for diagnostic criteria. *J Clin Epidemiol*. 2003;56(12):1150-1156. doi:10.1016/S0895-4356(03)00211-7

20. Jünger S, Payne SA, Brine J, Radbruch L, Brearley SG. Guidance on Conducting and Reporting Delphi Studies (CREDES) in palliative care: recommendations based on a methodological systematic review. *Palliat Med*. 2017;31(8):684-706. doi:10.1177/0269216317690685

21. Diamond IR, Grant RC, Feldman BM, et al. Defining consensus: a systematic review recommends methodologic criteria for reporting of Delphi studies. *J Clin Epidemiol*. 2014;67(4):401-409. doi:10.1016/j.jclinepi.2013.12.002

22. Coenen M, Kuijf HJ, Huenges Wajer IMC, et al; Alzheimer's Disease Neuroimaging Initiative. Strategic white matter hyperintensity locations for cognitive impairment: a multicenter lesion-symptom mapping study in 3525 memory clinic patients. *Alzheimers Dement*. 2023;19(6):2420-2432. doi:10.1002/alz.12827

23. Charidimou A, Boulouis G, Frosch MP, et al. The Boston criteria version 2.0 for cerebral amyloid angiopathy: a multicenter, retrospective, MRI-neuropathology diagnostic accuracy study. *Lancet Neurol*. 2022;21(8):714-725. doi:10.1016/S1474-4422(22)00208-3

24. McKeith IG, Boeve BF, Dickson DW, et al. Diagnosis and management of dementia with Lewy bodies: fourth consensus report of the DLB consortium. *Neurology*. 2017;89(1):88-100. doi:10.1212/WNL.0000000000004058

25. McKeith IG, Ferman TJ, Thomas AJ, et al; prodromal DLB Diagnostic Study Group. Research criteria for the diagnosis of prodromal dementia with Lewy bodies. *Neurology*. 2020;94(17):743-755. doi:10.1212/WNL.0000000000009323

26. Jack CR Jr, Andrews JS, Beach TG, et al. Revised criteria for diagnosis and staging of Alzheimer disease: Alzheimer's Association Workgroup. *Alzheimers Dement*. 2024;20(8):5143-5169. doi:10.1002/alz.13859

27. Dubois B, Villain N, Schneider L, et al. Alzheimer disease as a clinical-biological construct—an international working group recommendation. *JAMA Neurol*. 2024;81(12):1304-1311. doi:10.1001/jamaneurol.2024.3770

28. Arnold MR, Coughlin DG, Brumbach BH, et al. α-Synuclein seed amplification in CSF and brain from patients with different brain distributions of pathological α-synuclein in the context of co-pathology and non-LBD diagnoses. *Ann Neurol*. 2022;92(4):650-662. doi:10.1002/ana.26453

29. Benussi A, Premi E, Grassi M, et al. Genetic Frontotemporal Dementia Initiative (GENFI). Diagnostic accuracy of research criteria for prodromal frontotemporal dementia. *Alzheimers Res Ther*. 2024;16(1):10. doi:10.1186/s13195-024-01383-1

30. Meinel TR, Wardlaw JM, Kent DM. Is incidentally discovered covert cerebrovascular disease ignorable? *JAMA Neurol*. 2024;81(5):437-438. doi:10.1001/jamaneurol.2023.5456

31. Fazekas F, Chawluk JB, Alavi A, Hurtig HI, Zimmerman RA. MR signal abnormalities at 1.5 T in Alzheimer dementia and normal aging. *AJR Am J Roentgenol*. 1987;149(2):351-356. doi:10.2214/ajr.149.2.351

32. Debette S, Schilling S, Duperron MG, Larsson SC, Markus HS. Clinical significance of magnetic resonance imaging markers of vascular brain injury: a systematic review and meta-analysis. *JAMA Neurol*. 2019;76(1):81-94. doi:10.1001/jamaneurol.2018.3122

33. Weaver NA, Kuijf HJ, Aben HP, et al. Strategic infarct locations for poststroke cognitive impairment: a pooled analysis of individual patient data from 12 acute ischemic stroke cohorts. *Lancet Neurol*. 2021;20(6):448-459. doi:10.1016/S1474-4422(21)00060-0

34. Weaver NA, Mamdani MH, Lim JS, et al. Disentangling poststroke cognitive deficits and their neuroanatomical correlates through combined multivariable and multioutcome lesion-symptom mapping. *Hum Brain Mapp*. 2023;44(6):2266-2278. doi:10.1002/hbm.26208

35. Iadecola C, Duering M, Hachinski V, et al. Vascular cognitive impairment and dementia: JACC scientific expert panel. *J Am Coll Cardiol*. 2019;73(25):3326-3344. doi:10.1016/j.jacc.2019.04.034

36. Lafosse JM, Reed BR, Mungas D, Sterling SB, Wahbeh H, Jagust WJ. Fluency and memory differences between ischemic vascular dementia and Alzheimer disease. *Neuropsychology*. 1997;11(4):514-522. doi:10.1037/0894-4105.11.4.514

37. Bentvelzen AC, Kochan NA, Sachdev PS; VasCog-2 Neuropsychological Consortium. Harmonized neuropsychological assessment for Vascular Cognitive disorders: a Delphi consensus study. *Alzheimers Dement*. 2025;20(suppl 3):e086300. doi:10.1002/alz.086300

38. Dubois B, Villain N, Schneider L, et al. Alzheimer disease as a clinical-biological construct—an international working group recommendation. *JAMA Neurol*. 2024;81(12):1304-1311. doi:10.1001/jamaneurol.2024.3770

39. Hosoki S, Hansra GK, Jayasena T, et al. Molecular biomarkers for vascular cognitive impairment and dementia. *Nat Rev Neurol*. 2023;19(12):737-753. doi:10.1038/s41582-023-00884-1

40. Dadar M, Mahmoud S, Zhernovaia M, Camicioli R, Maranzano J, Duchesne S; CCNA Group. White matter hyperintensity distribution differences in aging and neurodegenerative disease cohorts. *Neuroimage Clin*. 2022;36:103204. doi:10.1016/j.nicl.2022.103204

41. Das RR, Seshadri S, Beiser AS, et al. Prevalence and correlates of silent cerebral infarcts in the Framingham offspring study. *Stroke*. 2008;39(11):2929-2935. doi:10.1161/STROKEAHA.108.516575

42. van den Berg E, Geerlings MI, Biessels GJ, Nederkoorn PJ, Kloppenborg RP. White matter hyperintensities and cognition in mild cognitive impairment and Alzheimer disease: a domain-specific meta-analysis. *J Alzheimers Dis*. 2018;63(2):515-527. doi:10.3233/JAD-170573

43. Staals J, Booth T, Morris Z, et al. Total MRI load of cerebral small vessel disease and cognitive ability in older people. *Neurobiol Aging*. 2015;36(10):2806-2811. doi:10.1016/j.neurobiolaging.2015.06.024

44. Mega Vascular Cognitive Impairment and Dementia (MEGAVCID) consortium. A genome-wide association meta-analysis of all-cause and vascular dementia. *Alzheimers Dement*. 2024;20(9):5973-5995. doi:10.1002/alz.14115

45. Greenberg SM, Aparicio HJ, Furie KL, et al; American Heart Association Stroke Council; Council on Cardiovascular and Stroke Nursing; and Council on Clinical Cardiology. Vascular neurology considerations for antiamyloid immunotherapy: a science advisory from the American Heart Association. *Stroke*. 2025;56(1):e30-e38. doi:10.1161/STR.0000000000000480

46. Smith EE, Barber P, Field TS, et al. Canadian Consensus Conference on Diagnosis and Treatment of Dementia (CCCDTD)5: guidelines for management of vascular cognitive impairment. *Alzheimers Dement (N Y)*. 2020;6(1):e12056. doi:10.1002/trc2.12056

47. Smith EE, Markus HS. New treatment approaches to modify the course of cerebral small vessel diseases. *Stroke*. 2020;51(1):38-46. doi:10.1161/STROKEAHA.119.024150

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